

Coaxial-Ceramic Resonator Filters and Multiplexers

50Ω

DC to 6 GHz

The Big Deal

- Low insertion loss with excellent power handling
- Passbands up to 6 GHz
- Fractional bandwidth from <1 to 25%
- Excellent temperature stability
- Rugged construction to handle demanding environmental conditions



Product Overview

Mini-Circuits' *Coaxial-Ceramic Resonator filters* offer low insertion loss in very small form factors, using ceramic material with high dielectric constant and superior Q factor. Bandpass and bandstop filters, diplexer and multiplexer designs can be constructed using this technology. Low insertion loss combined with excellent power handling makes these filters well suited for transmitter and receiver signal chains. Advanced filter design and construction can achieve stopband width greater than 3x the center frequency

All our coaxial-ceramic resonator filters are built with rugged construction, qualified to withstand multiple demanding reflow cycles. Excellent repeatability across units is achieved through precise tuning and process control.

Key Features

Feature	Advantages
Low insertion loss	Low signal loss results in better SNR in signal chain
Fast roll-off	Higher selectivity results in better adjacent channel rejection and dynamic range
Wide stop band	Wide spur-free stopband results in better receiver sensitivity
Excellent power handling	Well suited for transmitter applications
Rugged Construction	These filter assemblies have been qualified over a wide range of thermal, mechanical and environmental conditions including withstanding the stress of extensive solder reflow cycles
Small Size	Very well suited for high performance applications where size is a constraint.
Temperature stability	Very minimal change in electrical performance across temperature makes these filters suitable for a wide range of operating conditions.

Notes

A. Performance and quality attributes and conditions not expressly stated in this specification document are intended to be excluded and do not form a part of this specification document.

B. Electrical specifications and performance data contained in this specification document are based on Mini-Circuit's applicable established test performance criteria and measurement instructions.

C. The parts covered by this specification document are subject to Mini-Circuits standard limited warranty and terms and conditions (collectively, "Standard Terms"); Purchasers of this part are entitled to the rights and benefits contained therein. For a full statement of the Standard Terms and the exclusive rights and remedies thereunder, please visit Mini-Circuits' website at www.minicircuits.com/MCLStore/terms.jsp



Bandpass Filter

50Ω 1320 to 1580 MHz

ZX75BP-1450-S+



Generic photo used for illustration purposes only

CASE STYLE: HY1238
Connectors Model
SMA-MF ZX75BP-1450-S+

Features

- Low insertion loss
- High selectivity
- Connectorized package

Applications

- Aeronautical navigation
- Radio astronomy
- Wireless medical telemetry
- Defense systems.

Electrical Specifications at 25°C

Parameter		F#	Frequency (MHz)	Min.	Typ.	Max.	Unit
Pass Band	Center Frequency	F1-F2	-	-	1450	-	MHz
	Insertion Loss	F1-F2	1320-1580	-	1.1	2.0	dB
	VSWR	F1-F2	1320-1580	-	1.6	1.92	:1
Stop Band, Lower	Insertion Loss	DC-F3	DC-1000	60	70	-	dB
		F3-F4	1000-1100	40	46	-	dB
	VSWR	DC-F4	DC - 1100	-	20	-	:1
Stop Band, Upper	Insertion Loss	F5-F6	2000-2150	40	54	-	dB
		F6-F7	2150-2500	60	75	-	dB
	VSWR	F5-F7	2000-2500	-	20	-	:1

Maximum Ratings

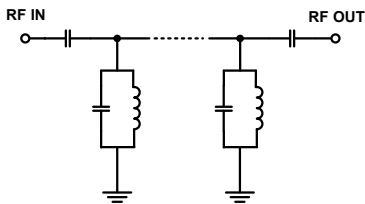
Operating Temperature	-40°C to 85°C
Storage Temperature	-55°C to 100°C
RF Power Input*	5 W max.

* Passband rating, derate linearly to 3.5W at 85°C ambient.
Permanent damage may occur if any of these limits are exceeded.

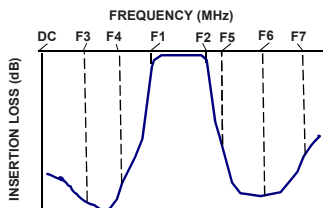
Typical Performance Data at 25°C

Frequency (MHz)	Insertion Loss (dB)	VSWR (:1)	Frequency (MHz)	Group Delay (nsec)
1	93.37	29644.67	1320	4.66
100	102.03	2935.22	1335	4.40
150	99.31	2206.70	1350	4.19
210	99.86	1129.66	1365	4.02
400	96.01	370.93	1380	3.87
600	96.77	172.86	1395	3.76
610	94.03	166.40	1410	3.67
1000	72.86	63.20	1425	3.59
1100	46.78	44.80	1440	3.53
1150	31.42	32.35	1450	3.49
1182	20.05	21.35	1470	3.44
1320	0.99	1.17	1485	3.42
1450	0.78	1.12	1500	3.43
1580	0.86	1.05	1515	3.46
1600	0.97	1.14	1530	3.52
1640	3.13	3.39	1545	3.60
1740	21.31	38.27	1560	3.71
2000	55.90	60.60	1565	3.76
2150	97.76	67.26	1570	3.81
2500	80.62	78.28	1580	3.96

Functional Schematic

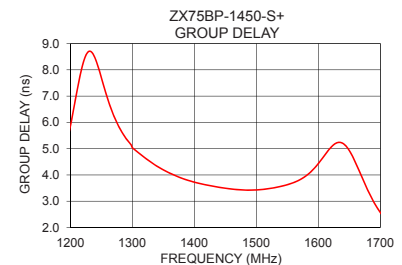
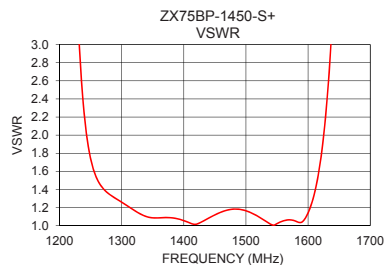
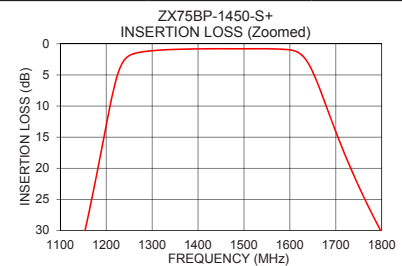
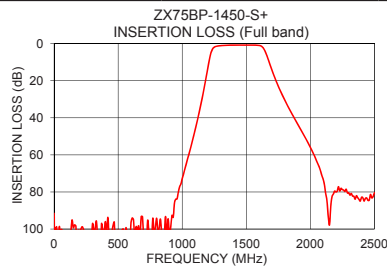


Typical Frequency Response



+RoHS Compliant

The +Suffix identifies RoHS Compliance. See our web site for RoHS Compliance methodologies and qualifications



Notes

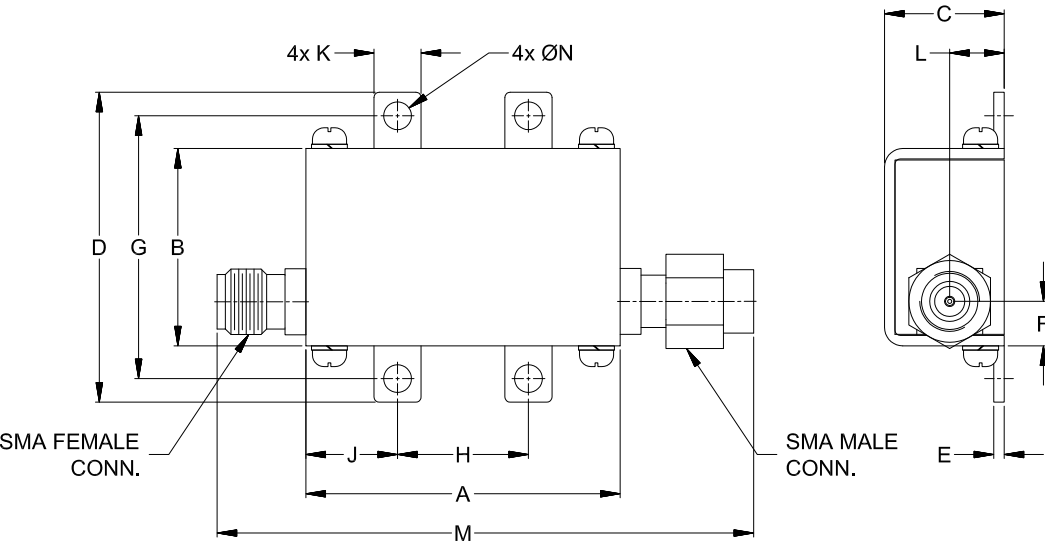
- A. Performance and quality attributes and conditions not expressly stated in this specification document are intended to be excluded and do not form a part of this specification document.
B. Electrical specifications and performance data contained in this specification document are based on Mini-Circuits' applicable established test performance criteria and measurement instructions.
C. The parts covered by this specification document are subject to Mini-Circuits standard limited warranty and terms and conditions (collectively, "Standard Terms"); Purchasers of this part are entitled to the rights and benefits contained therein. For a full statement of the Standard Terms and the exclusive rights and remedies thereunder, please visit Mini-Circuits' website at www.minicircuits.com/MCLStore/terms.jsp



Coaxial Connections

PORT - 1	SMA-MALE
PORT - 2	SMA-FEMALE

Outline Drawing



Outline Dimensions (^{inch}/_{mm})

A	B	C	D	E	F	G
1.20	.75	.46	1.18	.04	.17	1.00
30.48	19.05	11.68	29.97	1.02	4.32	25.40
H	J	K	L	M	N	Wt.
.50	.35	.18	.21	2.05	.106	grams
12.70	8.89	4.57	5.28	52.07	2.69	35.0

Note: Please refer to case style drawing for details

Notes
A. Performance and quality attributes and conditions not expressly stated in this specification document are intended to be excluded and do not form a part of this specification document.
B. Electrical specifications and performance data contained in this specification document are based on Mini-Circuit's applicable established test performance criteria and measurement instructions.
C. The parts covered by this specification document are subject to Mini-Circuits standard limited warranty and terms and conditions (collectively, "Standard Terms"); Purchasers of this part are entitled to the rights and benefits contained therein. For a full statement of the Standard Terms and the exclusive rights and remedies thereunder, please visit Mini-Circuits' website at www.minicircuits.com/MCLStore/terms.jsp